FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

E-COMMERCE PROJECT INITIATIVE



Project Initiative #3

- Financial Analysis and Assessment

Project Initiative #3

F

Financial Analysis and Assessment

อธิบายการแนวทางทำ Financial Modeling **รูปแบบของ** Outcome **และการประเมินทางการลงทุน เบื้องต้นสำหรับธุรกิจ** Startups

Financial Analysis & Assessment

อธิบายการแนวทางทำ Financial Modeling **รูปแบบของ** Outcome **และการประเมินทางการเงินสำหรับธุรกิจ** Startups



E-Commerce Project Initiative #3

Financial Projections for E-Commerce

- One of the biggest factors of a startup’s success is a thorough business plan that includes reliable financial projections for startups.
- The more accurate financial projections you make, the higher is the probability of attracting investments



Main Purpose of Making Financial Projection

Internal Financial Planning

Financial projections help you plan your startup budget, assess when you can expect the business to become profitable, and set benchmarks for achieving financial goals.

Seek External Funding

When you are in this stage, financial projections will help you in convincing potential investors that your startup will not only be profitable but also will have a good return on investment.

Two main approaches

There are two main methods of making financial projections for startups – top-down forecasting and bottom-up forecasting.



1. Using the top-down approach, you work from a macro/outside-in perspective towards a micro view. Typically industry estimates are taken as a starting point and narrowed down into targets that fit your startup.

2. With the bottom-up approach, you estimate expenses, costs, revenues, and investments based on the research and assumptions. The pitfall of this method is that it might fail to show the confidence needed to convince others of your startup potential.



Top-down Forecasting

Total Available Market (TAM)

The total worldwide market for a product or service

Serviceable Available Market (SAM)

The part of that market you address with your specific offering (the niche market) adjusted for your geographical reach

Serviceable Obtainable Market (SOM)

The part of SAM you can actually realistically capture given the existing competition. SOM is therefore equal to your sales target as it represents the value of the market share you aim to capture.



Bottom-up Forecasting

The bottom up approach

The bottom up approach is less dependent on external factors (the market), but leverages internal company specific data such as sales data or your company’s internal capacity. Contrary to the top down method, the bottom up approach begins with a micro/inside-out view and builds towards a macro view. This means a projection is made based on the main value drivers of your business.

How to make realistic assumption

- No matter what approach you choose to build your startup’s financial model, it is essential you can prove your numbers with realistic assumptions.
- The main problem is that previous data is not available, so you need to find a way to present the proof behind your estimated numbers.
- When you start discussing with investors, they are interested in knowing the argumentation behind your numbers. They are considering putting money in your startup, so you do not want to give them the feeling you are selling baloney!

How to make realistic assumption

- Assumptions can be anything that validates your numbers:
  - Market research
  - Web search volume
  - Contracts with suppliers
  - Pricing validation
  - Conversion rates

Startup's Financial Model

Financial projections combine three financial statements that are necessary to better understand the financial performance of your startup.



Outcomes of Startup's Financial Model

Income Statement

It is also known as a profit & loss statement and it focuses on the startup revenue and expenses generated during a particular period.

Cash flow statement

A cash flow statement aids you in understanding how startup operations will run.

Balance Sheet

A balance sheet presents a general overview of your startup’s financial health. It includes assets, liabilities, and owners’ equity in a specific period of time.

Income Statement

The four estimates included in the income statement are:

- Revenue
- Expenses
- Gains
- Losses

In the first year, you need to create a monthly income statement.

The second year, quarterly statements, and in the years to come, you will just need an annual income statement

COMPANY B
INCOME STATEMENT

For Year Ended September 28, 2019 (In thousands)

NET SALES	$	4,358,100
COST OF SALES		2,738,714
GROSS PROFIT		**1,619,386**
SELLING AND OPERATING EXPENSES		560,430
GENERAL AND ADMINISTRATIVE EXPENSES		293,729
TOTAL OPERATING EXPENSES		**854,159**
OPERATING INCOME		**765,227**
OTHER INCOME		960
GAIN (LOSS) ON FINANCIAL INSTRUMENTS		5,513
(LOSS) GAIN ON FOREIGN CURRENCY		(12,649)
INTEREST EXPENSE		(18,177)
INCOME BEFORE TAXES		**740,874**
INCOME TAX EXPENSE		257,642
NET INCOME	$	**483,232**

Source : Harvard Business School Online

Cash Flow Statement

It will show more detail on how much money will flow into and out of your startup business in income and expenses.

- Operating activities include any changes made in cash flow, accounts receivable, inventory, depreciation, and accounts payable.
- Investing activities include cash inflows for sales of assets and cash outflows for any fixed assets, such as property and equipment.
- Financing activities show a business's cash sources from either investors or banks and expenditures of cash paid to shareholders.

[Company Name]

© Corporate Finance Institute®. All rights reserved.

Cash Flow Statement
[USD $ millions]

	2014	2015	2016	2017	2018
Operating Cash Flow					
Net Earnings	2,474	11,791	21,075	26,713	28,227
Plus: Depreciation & Amortization	19,500	18,150	17,205	16,544	16,080
Less: Changes in Working Capital	9,003	1,702	775	903	827
Cash from Operations	**12,971**	**28,239**	**37,505**	**42,354**	**43,480**
Investing Cash Flow					
Investments in Property & Equipment	15,000	15,000	15,000	15,000	15,000
Cash from Investing	**15,000**	**15,000**	**15,000**	**15,000**	**15,000**
Financing Cash Flow					
Issuance (repayment) of debt	-	-	(20,000)	-	-
Issuance (repayment) of equity	170,000	-	-	-	-
Cash from Financing	**170,000**	**-**	**(20,000)**	**-**	**-**
Net Increase (decrease) in Cash	167,971	13,239	2,505	27,354	28,480
Opening Cash Balance	-	167,971	181,210	183,715	211,069
Closing Cash Balance	**167,971**	**181,210**	**183,715**	**211,069**	**239,550**

Source : Corporate Finance Institute

Balance Sheet

Balance sheet has two parts. You have assets on one side and liabilities and owner's equity on the other side. Projecting three years into the future could make it possible for you to estimate a break-even point.

- Assets are resources with a financial value that a startup business owns and estimated to provide a future benefit. That includes generating cash flow, enhancing sales and reducing expenses. Assets usually include property, inventory and cash.
- Liabilities are, in general, obligations to the third party. They are monthly expenses that occur during business operations. It usually includes accounts payable and loans.
- Owners' equity is any amount left over after paying all liabilities. It's usually categorized as retained earnings: the sum of all net income earned minus all paid earnings since inception.

Balance sheet example

TEDDY FAB INC.
BALANCE SHEET
December 31, 2100

ASSETS	
Current assets	
Cash and cash equivalents	$ 100,000
Accounts receivable	20,000
Inventory	15,000
Prepaid expense	4,000
Investments	10,000
Total current assets	149,000
Property and equipment	
Land	24,300
Buildings and improvements	250,000
Equipment	50,000
Less accumulated depreciation	(5,000)
Other assets	
Intangible assets	4,000
Less accumulated amortization	(200)
Total assets	$ 472,100

LIABILITIES AND SHAREHOLDERS' EQUITY	
Current liabilities	
Accounts payable	$ 30,000
Notes payable	10,000
Accrued expenses	5,000
Deferred revenue	2,000
Total current liabilities	47,000
Long-term debt	200,000
Total liabilities	247,000
Shareholders' Equity	
Common stock	10,000
Additional paid-in capital	20,000
Retained earnings	197,100
Treasury stock	(2,000)
Total liabilities and shareholders' equity	$ 472,100

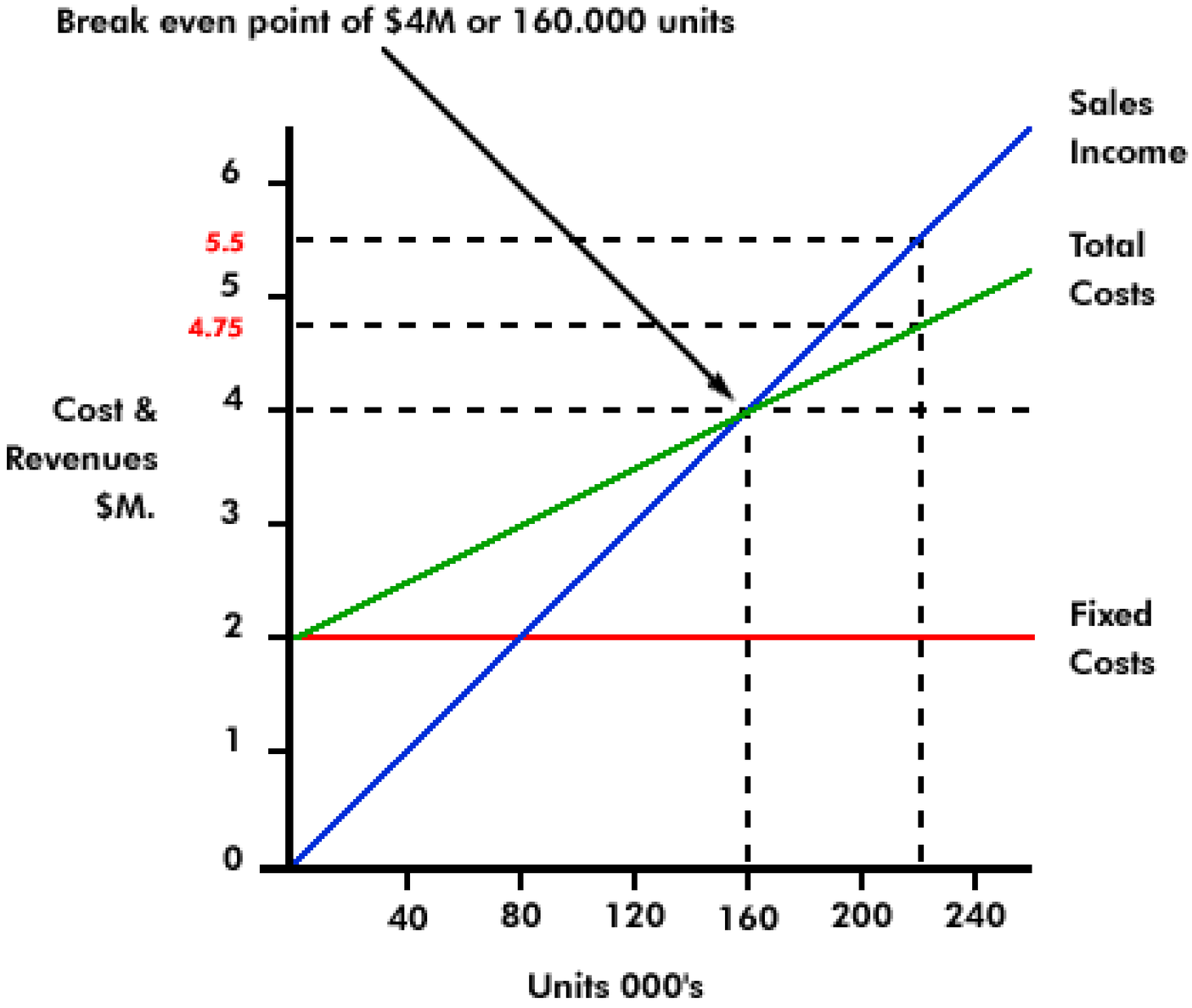
Source : Central Africa Tax Guide

Break-Even Analysis

- A break-even analysis shows your investor that you know the point at which you will start making a profit or the price that will cover your fixed costs.
- The break-even analysis is primarily for businesses making or selling products, or to set the right price for a product or service

Break-Even Analysis

It's usually shown as a graph with sales volume on the X axis and revenue on the Y axis. Then fixed an variable costs (those you must pay) are included. The break-even point marks the place where costs are covered.



Source : RMIT

Return on Investment (ROI)

- Return on investment (ROI) is a performance measure used to evaluate the efficiency or profitability of an investment or compare the efficiency of a number of different investments.
- ROI tries to directly measure the amount of return on a particular investment, relative to the investment’s cost.

Return on Investment

To calculate ROI, the benefit (or return) of an investment is divided by the cost of the investment. The result is expressed as a percentage or a ratio.

$$\text{ROI} = \frac{\text{Net Return on Investment}}{\text{Cost of Investment}} \times 100\%$$

ROI calculator

name Freitag

RETURN ON INVESTMENT

122%

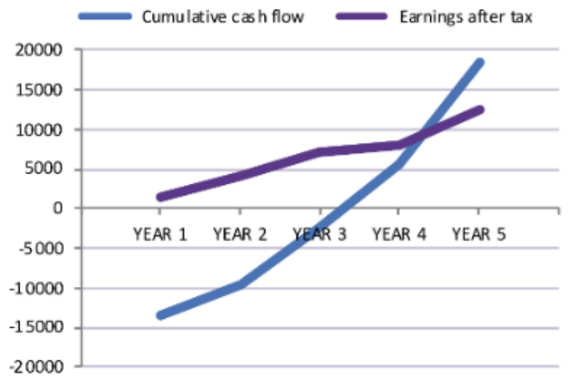
PAYBACK PERIOD

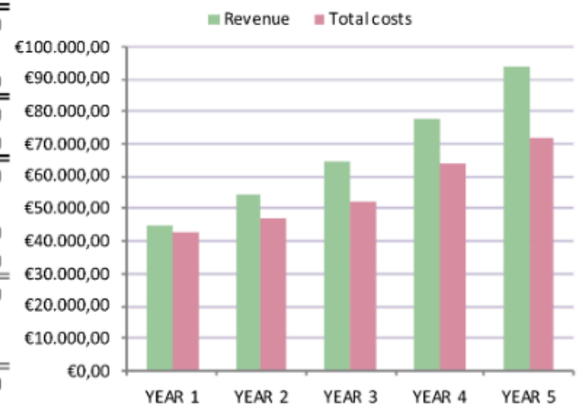
3,3 year

INITIAL INVESTMENT € 15.000,00

TAX 42%

	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5
Products sold	€500,00	€600,00	€720,00	€864,00	€1.037,00
Price	€90,00	€90,00	€90,00	€90,00	€90,00
Revenue	€45.000,00	€54.000,00	€64.800,00	€77.760,00	€93.330,00
Variable cost per product	€45,00	€45,00	€45,00	€45,00	€45,00
Total variable cost	€22.500,00	€27.000,00	€32.400,00	€38.880,00	€46.665,00
Fixed costs	€20.000,00	€20.000,00	€20.000,00	€25.000,00	€25.000,00
Total costs	€42.500,00	€47.000,00	€52.400,00	€63.880,00	€71.665,00
Earnings before tax	€2.500,00	€7.000,00	€12.400,00	€13.880,00	€21.665,00
Tax	€1.050,00	€2.940,00	€5.208,00	€5.829,60	€9.099,30
Earnings after tax	€1.450,00	€4.060,00	€7.192,00	€8.050,40	€12.565,70
Initial investment	-€15.000,00				
CUMULATIVE CASH FLOW	**-€13.550,00**	**-€9.490,00**	**-€2.298,00**	**€5.752,40**	**€18.318,10**





date

Source : Business Makeover

DON'T FORGET TO UPDATE

and review your financial projection with the actual figures.




Q&A
GO

Next Week Topics

Project Initiative #4:

- Minimal Viable Products

